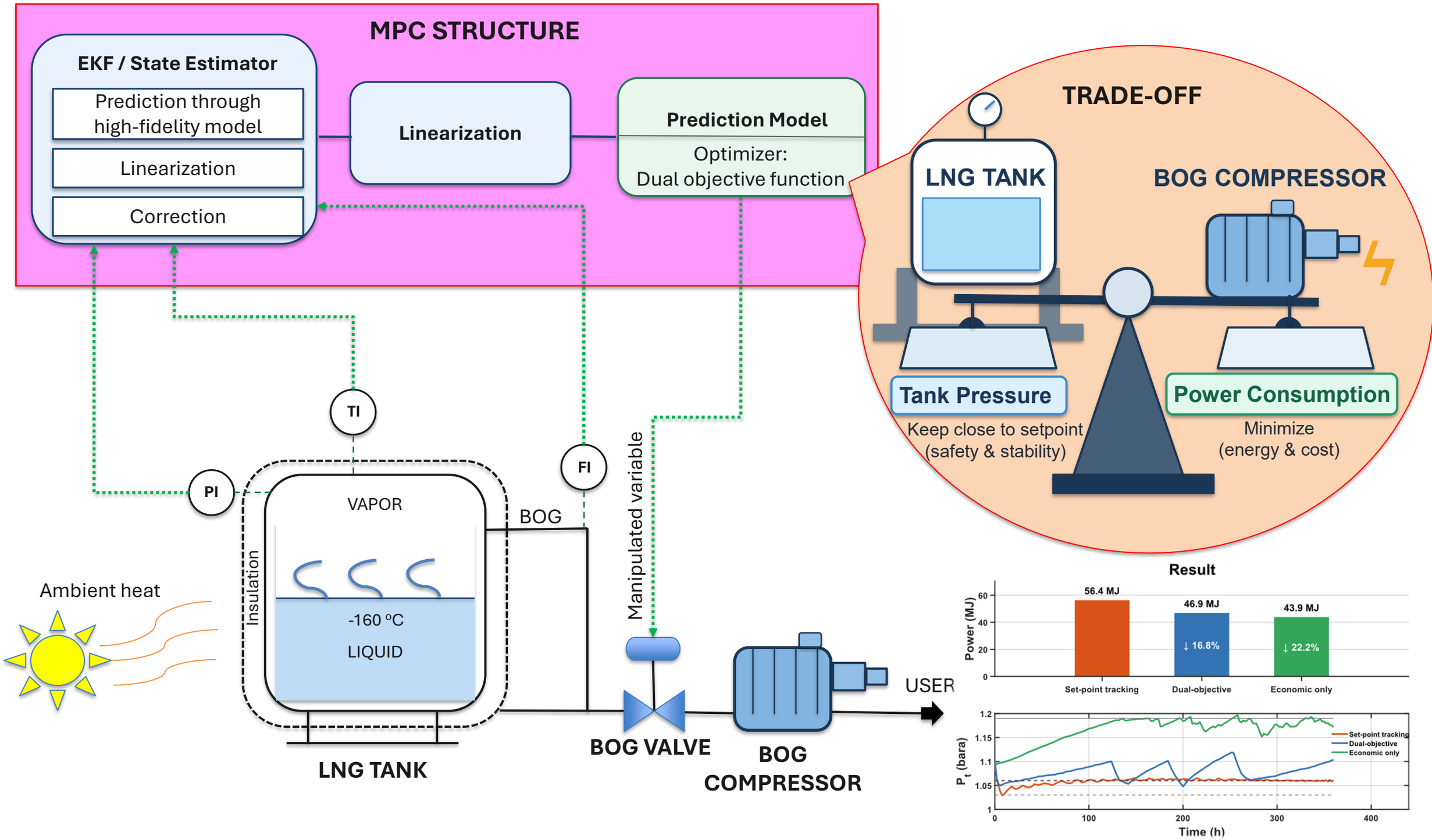

MPC STRUCTURE
EKF / State Estimator
Prediction through high-fidelity model
Linearization
Correction
Linearization
Prediction Model
Optimizer: Dual objective function
TRADE-OFF
LNG TANK
BOG COMPRESSOR
Tank Pressure
Keep close to setpoint (safety & stability)
Power Consumption
Minimize (energy & cost)
TI
PI
FI
VAPOR
BOG
Insulation
-160 °C
LIQUID
Manipulated variable
Ambient heat
LNG TANK
BOG VALVE
BOG COMPRESSOR
USER
Result
Power (MJ)
56.4 MJ
46.9 MJ
43.9 MJ
↓ 16.8%
↓ 22.2%
Set-point tracking
Dual-objective
Economic only
P_t (bara)
Time (h)